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Q1. Arrays, Array Applications and Structures**CODE**

```
#include <stdio.h>
#include <string.h>

char first(const char* s){
    int freq[256]={0};
    for(int i=0; s[i] != '\0'; i++){
        unsigned char index = (unsigned char)s[i];
        freq[index]++;
    }
    for(int i=0; s[i]!='\0';i++){
        unsigned char index =(unsigned char)s[i];
        if(freq[index]==1){
            return s[i];
        }
    }
    return '\0';
}

int main(){
    char msg[]="swiss";
    char result = first(msg);
    if (result!='\0'){
        printf("Firt non repeating character: %c\n", result);
    }else{
        printf("No Non repeating character found.\n");
    }
    return 0;
}
```

OUTPUT

(no output)

Q2. Arrays, Array Applications and Structures**CODE**

```
#include<stdio.h>
int search(int arr[], int size, int target){
    int low=0;
    int high =size-1;
    while (low≤high){
        int mid = low+(high-low)/2;
        if(arr[mid]==target){
            return mid;
        }
        if (arr[low]≤arr[mid]){
            if (target≥arr[low] && target < arr[mid]){
                high = mid-1;
            }else{
                low=mid+1;
            }
        }
        else{
            if (target>arr[mid]&& target≤arr[high]){
                low=mid+1;
            }else{
                high = mid-1;
            }
        }
    }
    return -1;
}

int main(){
    int arr[] = {4,5,6,7,0,1,2};
    int size = sizeof(arr)/sizeof(arr[0]);
    int target = 6;
    int result = search(arr, size, target);
    if(result≠-1){
        printf("Package is id %d found at index %d \n", target, result);
    }else{
        printf("package Id %d not found\n", target);
    }
    return 0;
}
```

OUTPUT

(no output)

Q3. Arrays, Array Applications and Structures**CODE**

```
#include<stdio.h>
struct Interval{
    int start;
    int end;
};
int main(){
    struct Interval arr[] = {{1,3},{2,6},{8,10},{15,18}};
    int n=sizeof(arr)/sizeof(arr[0]);
    for(int i=0;i<n-1;i++){
        for(int j=0;j<n-i-1;j++){
            if(arr[j].start >arr[j+1].start){
                struct Interval temp=arr[j];
                arr[j] =arr[j+1];

            }
        }
    }
    int index=0;
    for(int i=1;i<n;i++){
        if(arr[index].end ≥ arr[i].start){
            if(arr[i].end>arr[index].end){
                arr[index].end = arr[i].end;
            }
        }else{
            index++;
            arr[index]=arr[i];
        }
    }
    for(int i=0;i ≤ index;i++){
        printf("%d %d\n", arr[i].start, arr[i].end);
    }
    return 0;
}
```

OUTPUT

(no output)

Q4. Arrays, Array Applications and Structures**CODE**

```
#include<stdio.h>
#include<string.h>
#include<stdlib.h>
int freq[256] = {0};

int compare(const void *a, const void *b){
    char chara= *(char *)a;
    char charb = *(char *)b;

    if(freq[(unsigned char)charb] != freq[(unsigned char)chara]){
        return freq[(unsigned char)charb]- freq[(unsigned char)chara];
    }
    return chara-charb;
}

int main(){
    char str[100]= "Welcome to C programming";
    int len=strlen(str);
    for(int i=0;i<len;i++){
        freq[(unsigned char)str[i]]++;
    }
    qsort(str,len,sizeof(char), compare);

    printf("Frequent First elements in the given array are: %s \n", str);
    return 0;
}
```

OUTPUT

(no output)